

HOLOGRAPHIC DISPLAYS: Making Their Way Into Everyday Devices



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Electronic display

Digital device development never stands still, and the next big thing in devices could be the integration of 3D holography. This area is attracting considerable interest from tech developers and researchers. Technical advances in control of electronic displays have taken the ease of operation to a whole new level and allowed safe, fast and easy updates.

Holographic displays

A holographic display uses coherent light, such as that created by laser, to create a three-dimensional (3D) image in space. The hologram seems to be based on principles seen in photography and, in essence, is a 3D photograph.

Photographs aim to capture an image through light that has passed through a lens and onto a sensor. Modern digital cameras typically contain a sensor known as charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS), which translates the captured light into a digital image.

Holograms differ from photographs in the image capture method. These aim to

capture additional image data by using a laser and splitting the beam to illuminate the subject from two different angles. The light reflected off the subject is captured on a holographic plate, which is similar to an undeveloped strip of film. When light strikes an object with an irregular surface, it bounces off at a huge variety of angles, so different aspects of the object are disclosed when it's viewed from different perspectives.

In a hologram, a beam of light passes through a so-called diffraction fringe, which bends the light so that it, too, emerges at a host of different angles. One way to produce holographic video is to create diffraction fringes from patterns displayed on an otherwise transparent screen. The problem with that approach is that the pixels of the diffraction pattern have to be as small as the wavelength of the light they're bending, and most display technologies don't happily shrink down that much. In another approach, holographic-video displays are created by adopting acousto-optic modulation, in which precisely engineered sound waves are sent through a piece of

Holographic displays use light refraction to create 3D images that can be viewed without special glasses. This makes them highly desirable for applications ranging from medicine to augmented reality